METABOLIC RATE AND ORGAN SIZE DURING GROWTH FROM INFANCY TO MATURITY AND DURING LATE GESTATION AND EARLY INFANCY

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ABSTRACT. During growth from late infancy to maturity basal metabolic rate (BMR) per kilogram body weight decreases. The sources of metabolic energy, the cells, comprise two distinct groups. One, the high energy-producing organ cell mass (brain, liver, heart, and kidney), accounts for more than 60% of BMR although the sum of weight of these organs, (ΣOW) is less than 6% of total body weight in adult males. The other, muscle mass, is the sources of less than a third total BMR although it comprises more than 40% of total body weight. The average organ metabolic rate of the four organs, the OMR per kilogram organ weight per day is more than 300 kilocalories, or some 15 to 25 times that of muscle. The decline in BMR per kilogram body weight observed during growth is shown to be due principally to a slower growth of the high energy-producing organ cell mass relative to total body weight. OMR is proportional to the organ cell mass. ΣOW, OMR per kilogram organ weight is relatively constant during most of postnatal growth.

From the seventh month of gestation into infancy BMR increases more rapidly than either body weight or ΣOW, distinguishing this period of growth as one when the capacity for energy production is increasing.

This analysis, when applied to clinical disorders, allows a distinction to be made between changes in BMR that result from changes in body composition from those resulting from a change in metabolic activity *per se*. *Pediatrics*, 47: 169, 1971, METABOLIC RATE DURING GROWTH, ORGAN SIZE DURING GROWTH, GESTATION, OXYGEN CONSUMPTION, BMR.

THE basal metabolic rate (BMR) of an infant after the first month of age is 55 kilocalories/kilogram/day, of an adult 25 kcal/kilogram/day. This decrease in BMR /kilogram with growth is in keeping with the general observation that metabolic activity/kilogram is lower in larger organisms.[1,2] In adult man it has been shown that as much as two thirds of metabolic energy is derived from 4 organ systems: brain, splanchnic area (liver), heart and kidney,[3-5] although the sum of the weights of these organs (ΣOW) is less than 6% of total body weight. Muscle, on the other hand, is the source of 25 to 30% of basal metabolic energy or oxygen consumption but comprises 40 to 50% of total body weight and more than 80% of total cell mass in adult man. Table I presents a distribution of the basal metabolic energy among the 5 different systems (brain, liver, heart, kidney and muscle). The organ metabolic rate (OMR)* is 15 to 25 times that of muscle and averages 327 kilocalories/kilogram OW/day.

In the earlier report from this laboratory[5] the pattern of increase in weight of these four organ systems plus lung in relation to increase in body weight during normal growth was similar to the pattern of increase in BMR. When allometric or log-log plots of the same growth relations were made, the slopes of the comparative growth rates of ΣOW upon body weight and of BMR upon body weight also were similar.

An hypothesis was developed which

* Normally, metabolic activity of an organ is expressed in terms of oxygen consumption (VO_2), as milliliters of O_2/100 gm organ weight/minute. In order to compare organ metabolic activity to BMR we have converted VO_2 to OMR, expressed in terms of kcal/kilogram organ weight (OW)/day using 4.9 kilocalories/L oxygen utilized, assuming an RQ of 0.9[10] or for muscle 4.7 because muscle RQ is 0.7.[4]

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TABLE I

VO_2, Organ Metabolic Rate (OMR/Kilogram Body Weight), Metabolic Rate and Percent BMR Derived From Each Organ System in Adult Man

Organ	VO_2 (ml/minute /100 gm)	Organ		OMR (kcal/day)	% BMR
		Metabolic Rate (kcal/kg OW/day)	Weight (kg)		
1) Brain[6]	4.2	296	1.40	414	23.3
2) Liver[7]	4.1	290	1.60	464	26.1
3) Heart[8]	8.6	606	0.30	182	10.2
4) Kidney[9]	5.5	388	0.30	116	7.1
Total 1–4	—	327*	3.60†	1,177	66.7
5) Muscle[4]	0.26	17.6	28.3	500	28.1
Total 1–5	—	—	—	1,677	94.2
70 kg man	—	25.5	—	1,780	100.0

* OMR/kg OW derived from total kilocalorie/day of 4 organ systems, 1,177 ÷ total weight (ΣOW) for the 4 systems—3.60 kg.

† Metabolic rate (kilocalorie/kg OW 1 day) = VO_2 (milliliter/minute/100 gm) × 10 × 1,440 minute/day ÷ 1,000 × 4.9 kcal/liter oxygen (assuming an R.Q. of 0.9).[10]

stated that during growth the total metabolic activity of an organ increases in proportion to organ size and that OMR/kilogram OW is relatively constant. This contrasts with the findings among species of different size where OMR/kilogram OW is lower in larger animals.[1,2,5] In a second report, evidence was presented that renal oxygen consumption, as derived from glomerular filtration rate, is proportional to kidney size during normal growth.[11]

The present report is an extension of the mathematical analysis of this hypothesis. Data are also presented from the literature comparing BMR or standard VO_2 to body size and organ size (ΣOW) during growth from the seventh to tenth months of gestation.

The evidence supports the hypothesis that OMR is proportional to organ size during growth from infancy to maturity. However, OMR in prematures of 7 months gestational age appears low and increases faster than organ size before reaching the OMR/kilogram OW that will be sustained through subsequent growth.

Part I: The Relation of BMR to ΣOW during Growth from Infancy (1 month after full-term birth) to Maturity

Material: Organ size at different ages was derived from Coppoletta and Wolbach[12] for children aged 1 month to 12 years and related to body weight for children beyond 12 years from Scammon's data taken from Boyd[13] and for adults from Rezek and Millard.[14] Data from Schulz, et al.[15] for infants was found to agree with that of Coppoletta and Wolbach. BMR at different ages was taken from Talbot.[16] Organ sizes were related to height, not weight. Accordingly, to correlate data of BMR to ΣOW, each were plotted against height (Fig. 1). Interpolated values at 10 cm height increments were obtained, and body weights appropriate for that height were taken.[17] The data so derived are in Table II.

To obtain muscle mass, creatinine excretion/kilogram body weight was averaged

TABLE II

Height, Weight, Sum of Organ Weight, and Basal Metabolic Rate of Male Humans From Infancy at Full-Term Birth to Maturity

Height (cm)	Weight* (kg)	ΣOW† (kg)	BMR‡ (kcal/day)
50	3.0	0.50	160
60§	5.5	0.85	300
70	8.0	1.25	450
80§	11.0	1.60	590
90	14.0	1.70	700
100	17.0	1.95	780
110§	19.0	2.15	830
120	23.0	2.25	930
130	25.0	2.45	1040
140§	31.0	2.60	1160
150	37.0	2.65	1300
160§	50.0	2.85	1480
170	62.0	3.10	1630
180§	71.0	3.65	1800

* Weight is 50 percentile weight/or height.[17]

† ΣOW = Brain + liver + heart + kidney. Data up to 140 cm from Coppoletta, and Wolbach[12] through 170 cm Boyd,[13] adult from Rizek and Millard.[14]

‡ BMR from Talbot.[16]

§ Used in Table III with muscle data.

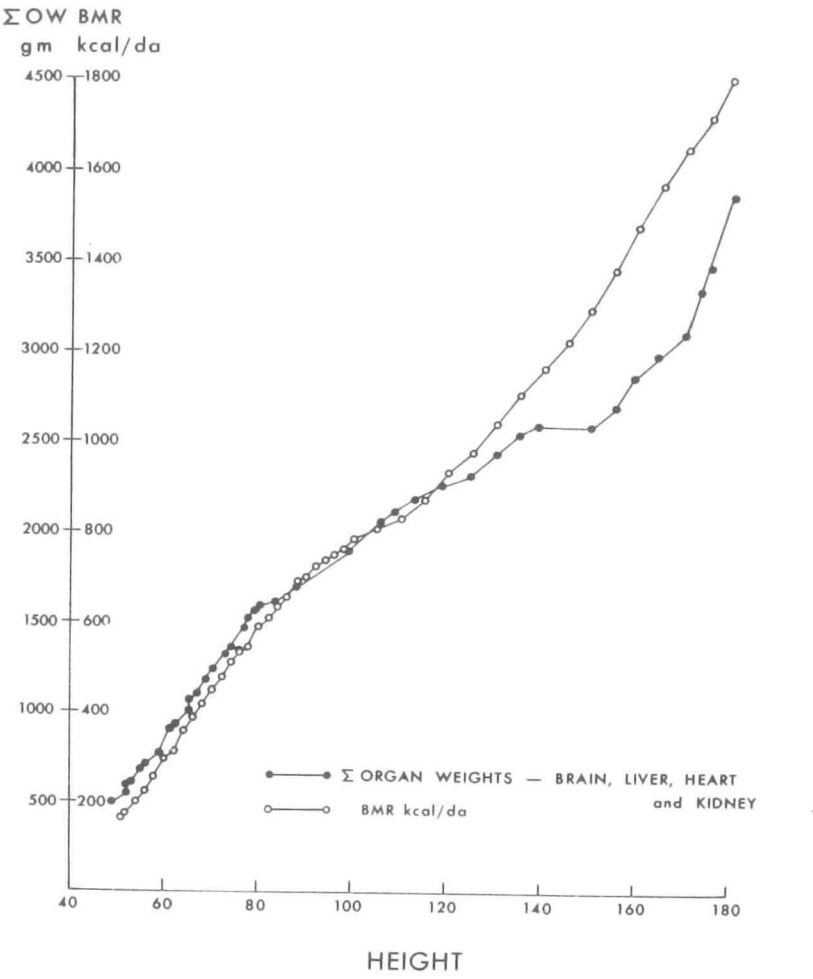


FIG. 1. The sum of weights of the 4 organs (brain, liver heart, and kidney (ΣOW) and basal metabolic rate (BMR) plotted against height during normal growth in humans. Data from Table II.

for children of several weight ranges (from the data recorded by Cheek[18]) and Picon-Reategui.[19] Applying the assumption that 1 gm creatinine is derived from 20 kg of muscle, muscle mass was calculated for 6 points in growth where the creatinine data were sufficient to derive muscle mass for body weight used in Table II. Muscle metabolic rate (MMR) reported by Andres, et al.[4] and the percent of total BMR derived from muscle was calculated (Table III). The BMR less MMR was obtained by difference and assumed to represent mostly the OMR (brain, liver, heart and kidney) (Table I).

Regression analyses were done by the method of least squares. Where the slope of a log-log plot was determined, the log of the values was used to calculate the regression.

RESULTS: The BMR and ΣOW (Fig. 1) are seen to increase in proportion to each other until a height of 120 cm (age 12) is reached. From that point, BMR increases more than ΣOW. When BMR and ΣOW are plotted against body weight (Fig. 2), a similar parallelism is noted, but the ΣOW increases less than BMR in the later stages of growth.

TABLE III

DERIVATION OF MUSCLE MASS, MUSCLE METABOLIC RATE AND THE PROPORTION OF BMR DERIVED FROM MUSCLE

Height (*cm*)	*Weight* (*kg*)	*Creatinine Index* (*mg/kg/day*)	*Muscle Mass* (*kg*)	*Muscle MR* (*kcal/day*)	*BMR* (*kcal/day*)	*OMR* (*kcal/day*)	*MMR/BMR* %	*ΣOW* *kg*
60	5.5	11.0	1.21	21	300	279	7.0	0.85
80	11.0	11.7	2.57	45	590	545	7.6	1.60
110	19.0	17.5	6.65	117	830	713	14.0	2.15
140	31.0	18.7	11.6	204	1,160	956	17.6	2.60
160	50.0	21.2	21.2	373	1,480	1,107	25.2	2.85
180	70.0	20.2	28.3	500	1,800	1,300	27.7	3.65

Creatinine index, milligram of creatinine excreted/kilogram body weight/day calculated from data of Cheek,[18] using the following weight ranges and average daily creatinine excretions/kilogram determined within that range: 4–7 kg, 11.0; 7–12 kg, 11.7; 19–25 kg, 17.5; 25–35 kg, 18.5; 35–60 kg, 21.8: Creatinine index for 70 kg adult man from Picon and Reatigui.[19]

Muscle mass in kilogram = creatinine index × .02 muscle/milligram creatinine/day × body weight in kilogram.

Muscle metabolic rate in kilocalorie/day = 2.6 ml oxygen/kilogram forearm/minute[4] × .0047 kilocalorie/milliliter oxygen (where respiratory quotient[10] is .7) × 1,440 minute/day = 17.6 kilocalorie/kilogram muscle/day × muscle mass.

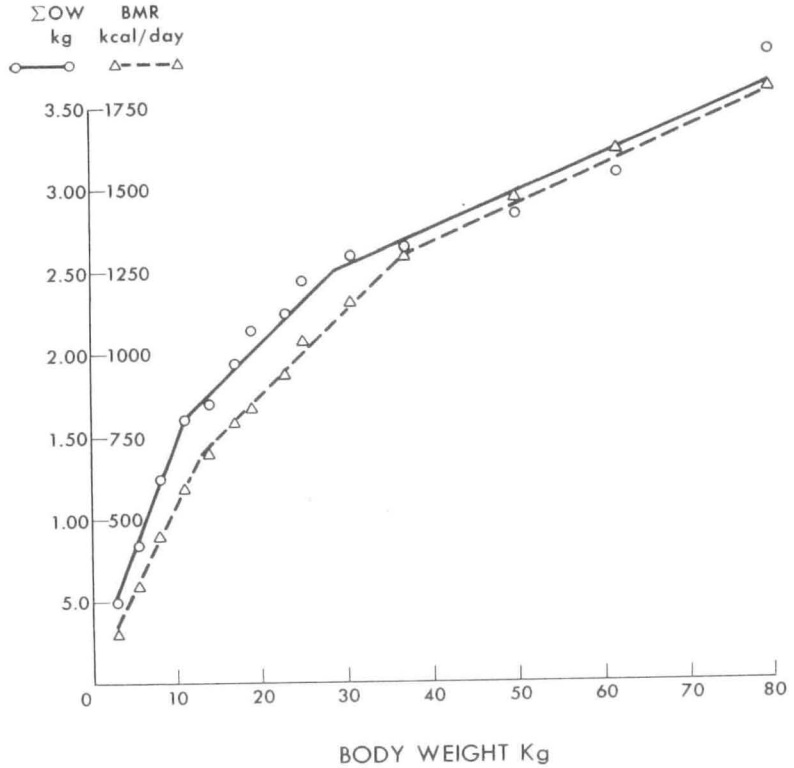


FIG. 2. ΣOW and BMR interpolated at 10 cm height increments from Figure 1 and plotted against body weight taken as the 50 percentile weight for height.[17] Data from Table II.

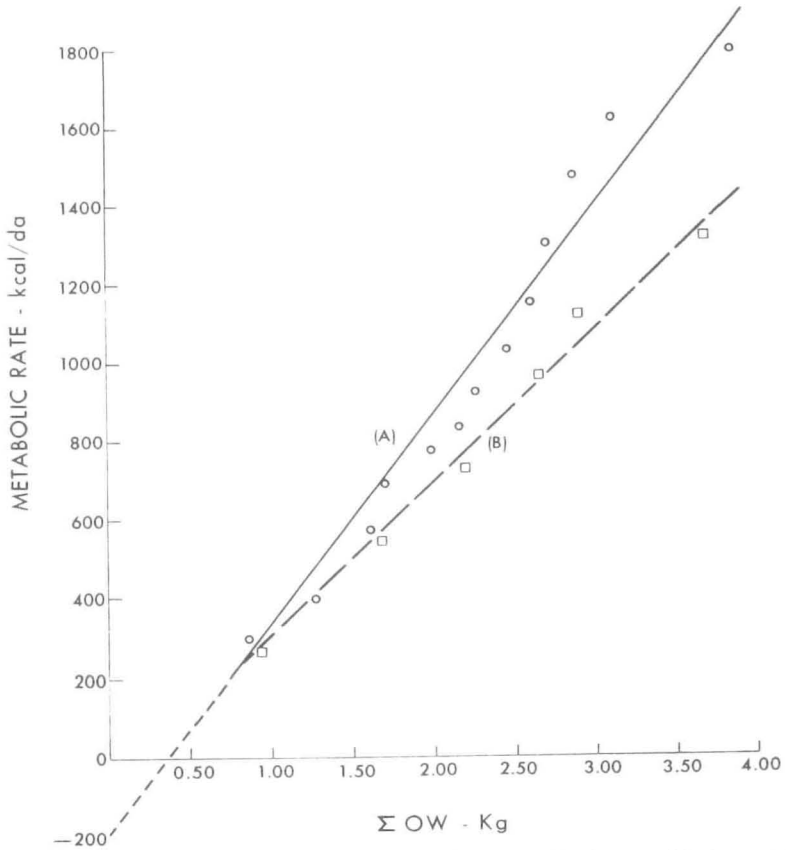


FIG. 3. BMR plotted against ΣOW (line A) using data from Table II; and BMR less muscle metabolic rate (MMR) which defines organ metabolic rate (OMR) plotted against ΣOW (line B) using data from Table III. Regression equations: Line (A) BMR (kcal/kg OW/da) = 555 · ΣOW (kg) − 232; r = 0.98 and Line (B) OMR (kcal/kg OW) = 381 · ΣOW (kg) − 53; r = 0.99.

A plot of BMR upon ΣOW (Fig. 3, line A) illustrates a close correlation of the two. However, by regression analysis the slope is 555, implying that OMR/kilogram OW/day is 555 kilocalories, a figure higher than the average observed (Table I) in adult man, 327 kilocalories.

An allometric plot (Fig. 4, line A) illustrates that the slope of the regression for all points is 1.22. One interpretation of this is that BMR is increasing at a greater rate than ΣOW, contrary to the original hypothesis.

A second alternative may be developed from the data in Table III.† Muscle contributes only 7% to BMR in the infant but as much as 25 to 30% in the adult. BMR less metabolic rate from muscle (MMR) should be nearly equal to the total OMR. If OMR is proportional to ΣOW throughout growth, then a plot of BMR less MMR or OMR de-

† The data from Table III used to calculate line B was taken from only 6 of the 14 points (Table II) used to calculate line A, because muscle mass could be calculated only for these 6 points. However, these 6 points are representative of all 14 points as evidenced by the fact that the slope of BMR upon ΣOW (line A) for these 6 points was 1.23 compared with a slope of 1.22 for all 14 points.

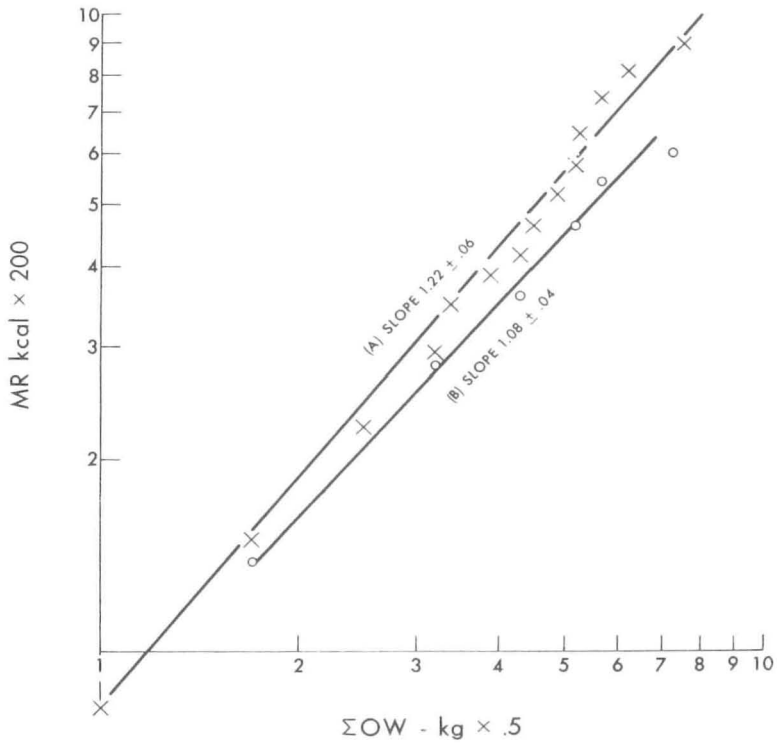


FIG. 4. A log-log plot of BMR against ΣOW (line A) and log-log plot of OMR against ΣOW (line B) using same data as in Figure 3. Slopes of lines of regression derived by method of least squares using logs of respective data. A slope of 1.0 implies proportional growth rates. The slope of line A, 1.22 is significantly different from 1.0; that of line B, 1.08, is not significantly different from 1.0

rived in this way against ΣOW during growth should describe a slope that approximates the average OMR/kilogram OW of adults. The slope 381 kilocalories/kilogram OW (Fig. 3, line B) compares with the average for adults of 327. The intercept (−53 kilocalories) brings these figures even closer to each other. A log-log plot of OMR against ΣOW should be 1.0 to fit the hypothesis. From the data of Table III plotted in Figure 4, line B the slope is 1.08 ± .04.

These findings lend considerable support to the general hypothesis that organ metabolic activity is proportional to organ mass during growth. The evidence is indirect and depends upon the assumption that MMR is also proportional to muscle mass during growth. A direct test of the hypothesis in the case of the kidney has been cited.[11] Kennedy and Sokoloff[6] reported brain oxygen consumption of 9 children ranging in age from 3 to 11 years. There was no change in VO_2/100 gm brain/minute which correlated with the age of the children. The mean of all children (5.17 ml O_2/100 gm brain/minute) was slightly higher than the mean they obtained in adults—4.18. The magnitude of the difference is small and well within the limits of confidence of the relations developed in the hypothesis.

Part II: The Relation of BMR or VO_2 to ΣOW during Growth from the Seventh to Tenth Month Post-Conception

MATERIAL: Sinclair, et al[20] reported the oxygen consumption of 92 infants ranging

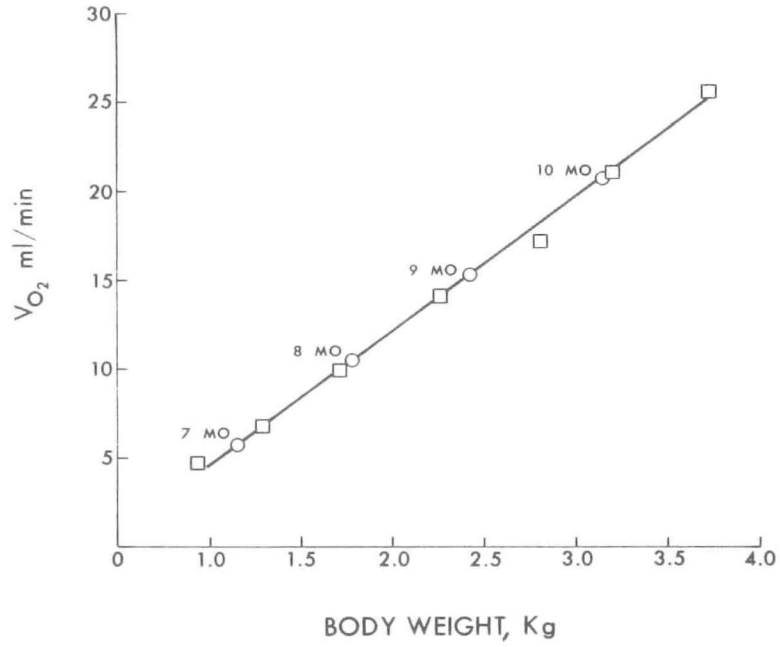


FIG. 5. Average VO_2 versus average body weight (□) of 7 weight groups derived from data of Sinclair, et al.[20] The VO_2 for infants of the weight for gestational age reported by Schulz, et al.[15] was interpolated (○) and used to calculate BMR in Table IV.

in weight from 730 gm to 3,840 gm who were normal in weight for gestational age. The average VO_2 of children in each of 7 weight intervals has been calculated and related to the average weight for each interval (Fig. 5). Schulz, et al.[15] recorded the average body weight and organ weights of infants at different gestational ages who died acutely in the first few days of life and who did not have any evidence of pathology affecting the body weight or organ weight. By interpolation from Figure 5 it was possible to compare VO_2 and ΣOW of infants of known weight at gestational ages of 7, 8, 9 and 10 months, respectively.

RESULTS: From these data VO_2 or BMR‡/kilogram body weight/day and ΣOW as percent body weight can be derived at different gestational ages. The results are in Table IV. BMR/kilogram increases from 37.4 to 45.8 kilocalories/kg. ΣOW as percent of body weight decreases from 21.0 to 18.5%. The divergent direction of these relations indicates that BMR increases at a faster rate than ΣOW.

The rate of growth of BMR compared with body weight and ΣOW for the infants in

TABLE IV

COMPARING BMR/KILOGRAM BODY WEIGHT OF INFANTS OF DIFFERENT GESTATIONAL AGE WITH ΣOW AS PERCENT BODY WEIGHT—INFANTS WERE NORMAL IN WEIGHT FOR GESTATIONAL AGE

Gestational Age	*Month*			
	7	*8*	*9*	*10*
Body weight (gm)	1,145	1,780	2,420	3,170
ΣOW (gm)	239	378	483	587
BMR (kcal/day)	42.4	74.1	106	146
BMR (kcal/kg/day)	37.4	41.6	43.7	45.8
ΣOW (% body weight)	21.0	21.2	20.0	18.5

BMR estimated from resting VO_2 as determined by Sinclair, et al.[20] and interpolated from Figure 5. Body weight and ΣOW for gestational age were those recorded by Schulz, et al.[15]

BMR (kilocalorie/day) = VO_2 (milliliter oxygen) × 1,440 minute/day ÷ 1,000 × 4.9 kcal/liter oxygen (assuming an RQ of 0.9).[10]

‡ BMR infants is not easily defined. A standard MR measured in a thermoneutral environment in resting infants was used as a relative measure of BMR or resting VO_2.[20]

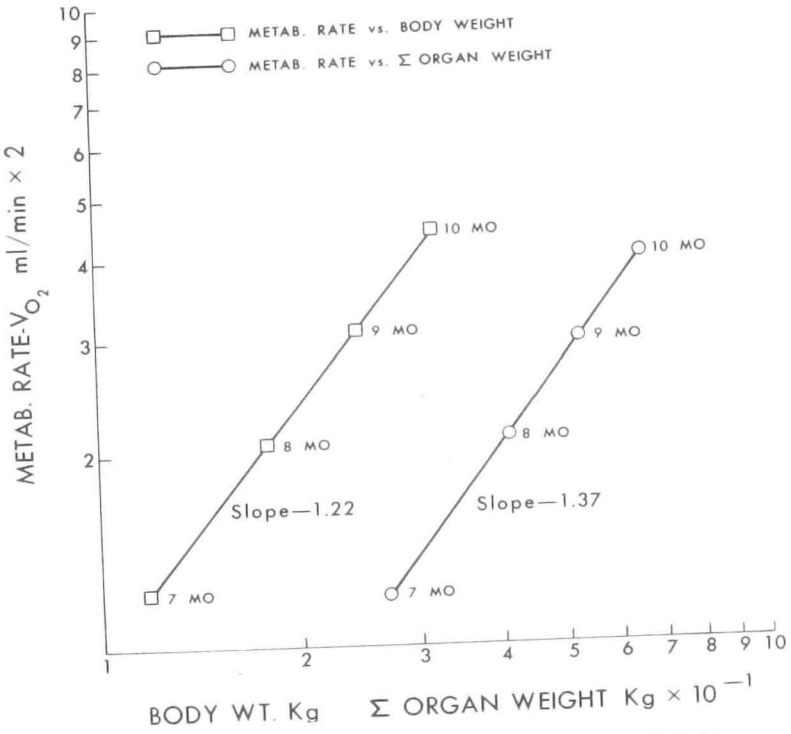


FIG. 6. Log-log plot of BMR versus body weight (line A) and BMR versus ΣOW (line B) for infants from the seventh to tenth month gestational age using data in Table IV. The slopes of lines A, 1.22, and line B, 1.37, derived by method of least squares—using logs of respective data—are significantly greater than 1.0.

this growth period is derived from the allometric plots of Figure 6. The slope of log VO_2 upon log body weight is 1.22 (line A), indicating that metabolic activity is increasing at a faster rate than body weight. The slope of log VO_2 upon log ΣOW is 1.37 (line B). Since, in this period of development, muscle mass is less than in the infant at birth (Table III) and contributes very little to BMR, the slopes relating VO_2 to body weight 1.22, and VO_2 to ΣOW—1.37 indicate a rate of increase in metabolic activity that is greater than the increase in body weight or ΣOW.

It is generally appreciated that the last period of intra-uterine growth and the early months of infancy are periods in which organ growth is accomplished to a large degree by cell division more than cell growth, while later the reverse is true. Data from the kidney[11] does not provide evidence that a change in this pattern is directly influential upon the ratio of organ metabolic rate to organ size. The earlier period of growth represented in the analysis of the infants from 7 to 10 months gestation, however, is presumed to represent a period of cell maturation and organization as well as cell multiplication.

We have drawn the conclusion that the early period of organ development is characterized by a low rate of oxygen consumption or metabolic activity and the capacity for oxygen consumption increases during this phase of development at a greater rate than organ size. At some point in early postnatal development in humans the pattern changes so that subsequent increases in organ oxygen consumption or metabolic activity are proportional to increases in organ

TABLE V

CALCULATION OF PERCENT BMR CONTRIBUTED BY BRAIN METABOLISM (BR MR KILOCALORIE/DAY), LIVER METABOLISM (LMR KILOCALORIE/DAY), AND MUSCLE METABOLISM (MMR KILOCALORIE/DAY)

Body Weight (kg)	*Total BMR*	*Brain Weight (kg)*	*BrMR (kcal/day)*	*Liver Weight (kg)*	*LMR (kcal/day)*	*Muscle Mass (kg)*	*MMR (kcal/day)*	*% Total BMR as*		
								BrMR	*LMR*	*MMR*
5.5	300	0.65	192	0.19	55	1.21	21	64	18	7
11.0	590	1.05	311	0.37	107	2.57	45	53	18	8
19.0	830	1.24	367	0.64	186	6.65	117	44	22	14
31.0	1,160	1.35	400	0.94	272	11.59	204	35	23	18
50.0	1,480	1.36	403	1.17	339	21.20	373	27	23	25
70.0	1,800	1.40	414	1.65	478	28.3	500	23	27	28

BrMR = 296; LMR = 290; MMR = 17.6 kilocalorie/kilogram/day (see Table I).

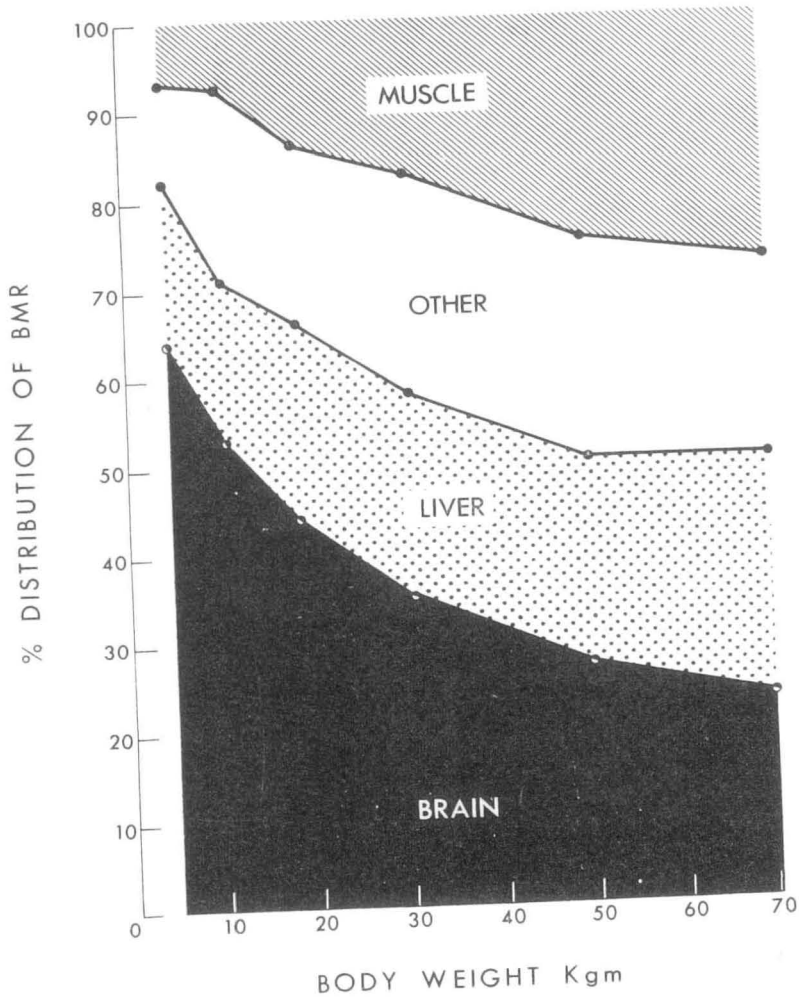


FIG. 7. The changing contribution to BMR derived from brain, liver, and muscle as percent total BMR as growth proceeds. Data derived from Table V.

mass. This applies to the brain, liver or splanchnic area, heart, and kidney. Any specific organ function in which oxygen consumption is not directly rate-limiting may exhibit a different pattern of development of function to size. However, those in which oxygen consumption is rate-limiting, e.g., sodium reabsorption from glomerular filtration rate, resting cardiac output should conform to the pattern of oxygen consumption or OMR/kg organ weight as noted herein.

SPECULATION

With the assumption that OMR is proportional to organ size one can partition the approximate distribution of energy production at different ages for normal children. Figure 7 illustrates the partition of BMR among brain, liver, and muscle at different ages. Changes in BMR or VO_2 observed with disease can be compared with these values as a first approximation toward evaluating whether the changes observed are due to differences in metabolism or to body composition.

One example may illustrate the point: BMR/kilogram body weight has been observed to be normal or low in children with malnutrition;[21] with refeeding, it increases to levels above normal. Montgomery[21] speculated that OMR may be low in marasmic children in whom organ size is disproportionately high. With refeeding, their OMR increases to normal and BMR/kilogram body weight becomes high. By using techniques to measure organ metabolic mass and muscle mass it should be possible to appraise these and other changes in metabolism with greater accuracy.

The hypothesis, if sustained, illustrates the hazard of using reference standards for BMR during growth which have been developed over a broad range in size for mature mammals. The ratio of OMR to organ weight during early growth (seventh to tenth month gestation) is the opposite of that seen generally; OMR/kilogram OW increases as body size increases. During growth from infancy to maturity, OMR/kilogram OW remains constant. Only among mature species of different size is OMR/kilogram OW less among larger animals.

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